

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 6: Global Jet Streams, Clear Air Turbulence & Tropopause

Syllabus Source: CAE Oxford (pp. 175-208) | Pass Target: $\geq 90\%$ AviationExam

1. JET STREAM ARCHITECTURE & CRITERIA

A Jet Stream is a flat, tubular current of quasi-horizontal wind characterized by strong vertical and lateral wind shears:

- ICAO Official Definition: A high-altitude wind stream with a minimum CORE SPEED of at least 60 KNOTS (typically 100 to 200+ knots)!
- Polar Front Jet (PFJ): Located along the Polar Front between polar and tropical air masses. Core height: $\sim 30,000$ ft (FL 300 / 300 hPa). Discontinuous, meanders from 40° to 60° latitude. Strongest in winter!
- Subtropical Jet (STJ): Located between Hadley and Ferrel cells at $\sim 30^\circ$ latitude. Core height: $\sim 40,000$ ft (FL 400 / 200 hPa). More stable in position, blowing from West to East.
- Easterly Tropical Jet: Summer jet in the Northern Hemisphere flowing from East to West over southern Asia and central Africa at FL 450 to FL 500.

2. CLEAR AIR TURBULENCE (CAT) DYNAMICS

CAT occurs in cloud-free air, primarily near jet streams and the tropopause break:

- Location of Maximum CAT: Severe CAT is concentrated on the COLD (POLAR) SIDE of the jet core and directly ABOVE the jet core, where horizontal and vertical shears are greatest!
- Critical Shear Criteria: Severe CAT is expected when: Vertical Shear ≥ 6 knots per 1,000 ft, OR Horizontal Shear ≥ 18 knots per 100 NM!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Finding Core and Escaping Severe CAT

Location of Maximum Clear Air Turbulence (CAT): - The most violent CAT is concentrated on the COLD POLAR SIDE (left side looking downwind) and ABOVE the jet core. - Vertical shear exceeds 6 kt / 1,000 ft and horizontal shear exceeds 20 kt / 100 NM. Cockpit In-Flight Action: - If OAT is dropping rapidly: You are penetrating the cold polar side -> CLIMB or turn toward warmer air. - If OAT is rising: You are entering the warm side -> DESCEND or turn south. - Immediately set Turbulence Penetration Speed (V_{RA} / M_{RA}).

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Evaluating Wind Shear for Severe CAT Encounter

SCENARIO (AviationExam High-Altitude Navigation Drill): An airliner is cruising at FL 350 on an airway near the Polar Front Jet: - At FL 350, wind is 270° at 140 knots - The pilot is cleared to descend to FL 310 - At FL 310 (4,000 ft below), wind is reported as 270° at 170 knots (near the jet core) QUESTION: What is the vertical wind shear per 1,000 ft, and should the crew anticipate severe Clear Air Turbulence (CAT)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Total Wind Speed Difference: - Wind at FL 310 = 170 knots - Wind at FL 350 = 140 knots - Difference in speed = 170 kt - 140 kt = 30 knots. Step 2: Calculate the Altitude Difference in Thousands of Feet: - Altitude difference = 35,000 ft - 31,000 ft = 4,000 ft = 4.0 thousand ft. Step 3: Calculate Vertical Wind Shear per 1,000 ft: - Vertical Shear = Speed Difference / Height Difference - Vertical Shear = 30 knots / 4.0 thousand ft = 7.5 knots per 1,000 ft! Step 4: Compare with the ICAO Severe CAT Threshold: - Threshold for Severe CAT = 6.0 knots per 1,000 ft. - Actual Vertical Shear = 7.5 knots per 1,000 ft. - Check: 7.5 kt/1,000 ft > 6.0 kt/1,000 ft -> EXCEEDS SEVERE THRESHOLD! FINAL ANSWER: Vertical shear is 7.5 kt/1,000 ft. Severe CAT is highly probable! The crew must switch on Fasten Seatbelt signs and select Turbulent Air Penetration Speed (V_{ra}).

[!] EXAM TRAP (AVIATIONEXAM) - Cold Side vs Warm Side CAT Trap

- The COLD (low pressure/polar) side of the jet stream has MUCH STRONGER SHEAR than the warm side! - If flying parallel to a westerly jet stream in the Northern Hemisphere: - North of the jet core = Cold side = MAXIMUM TURBULENCE. - South of the jet core = Warm side = Moderate/light turbulence.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the official ICAO minimum core wind speed required for a high-altitude wind current to be classified as a jet stream?

- [A]: 40 knots
- [B]: AT LEAST 60 KNOTS
- [C]: 100 knots
- [D]: 150 knots

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under WMO and ICAO definitions, a jet stream must have a minimum wind speed of 60 knots along its central axis.

Q2: In relation to the jet stream core, where is the most severe Clear Air Turbulence (CAT) typically located?

- [A]: Exactly inside the core where speed is fastest
- [B]: On the COLD (polar) side of the core and just above the core, where wind shear is greatest
- [C]: On the warm side below the core
- [D]: Equatorward of the jet

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Turbulence is generated by wind shear (speed gradient), not absolute speed. The polar side exhibits much tighter horizontal and vertical gradients.

Q3: What seasonal change occurs in the strength and geographic position of the Polar Front Jet (PFJ)?

- [A]: Strongest in summer and moves equatorward
- [B]: STRONGEST IN WINTER and moves EQUATORWARD (lower latitudes)
- [C]: Strongest in summer over the poles
- [D]: Unchanged throughout the year

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. In winter, the equator-to-pole temperature contrast is at its maximum, intensifying the pressure gradient and driving the jet stream further south with higher core speeds.